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Information-Matched Auditing of Conversational Emotion-Shift Forecasting

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ABSTRACT Emotion-shift forecasting predicts whether the next conversational utterance will express a different emotion before that utterance is observed. Comparisons are easily distorted when a learned model receives rich features but its transition baseline is given a noisy or unavailable current-emotion label, when hyperparameter budgets differ, or when target-turn information enters the representation. We present an information-matched, leakage-safe audit with an explicit commitment point, executable causality tests, equal four-configuration validation searches, and seed–dialogue-aware inference. Six causal model families are evaluated on four text corpora and two feature-source sensitivity configurations. In the deployable condition, a train-only current emotion classifier feeds the transition baseline; in the oracle diagnostic, both learned models and the baseline receive the same gold current emotion. Under the deployable condition, learned models improve over the predicted-label transition baseline on all four primary text corpora; DailyDialog gains are large and survive a 72-comparison Holm correction (best ΔROC–AUC $= +0.167$, adjusted $p = .0144$). Under matched gold information, the transition baseline remains competitive on IEMOCAP and is significantly stronger on MELD, whereas every model retains a large DailyDialog advantage (best $\Delta = +0.132$, adjusted $p = .0144$). We also report precision–recall AUC, calibration, a strictly utterance-local feature control, same-person self-shift forecasting, and a same-architecture future-access dose response. The results replace a binary "model versus inertia" narrative with a conditional one: conclusions depend materially on current-label availability, corpus construction, target semantics, feature provenance, and leakage discipline. Code, raw seed predictions, tests, and reporting artifacts support reproducible auditing.

INDEX TERMS emotion forecasting, conversational emotion recognition, evaluation protocol, data leakage, benchmarking

I. INTRODUCTION

Consider a call-center assistant that flags, one turn before it happens, that a customer is about to escalate from neutral to angry – giving the agent seconds to change course. This is the promise of *forecasting* emotion in conversation, as opposed to *recognizing* it: the target state does not yet exist when the prediction is made. It is also precisely the setting in which a model can look right without being right. If any feature – a pooled dialogue embedding, a bidirectional encoder, a context window one utterance too wide – depends even slightly on the turn being forecast, the model is not anticipating the future; it is reading it off a crib sheet built into its own inputs. The failure is silent because nothing in a standard run signals it happened: loss curves look normal, F1 looks respectable, the leaderboard entry looks like progress.

Emotion recognition in conversation (ERC) has a large, mature literature (§II). A smaller but growing thread reframes the task as forecasting – predicting an upcoming emotional shift before the triggering utterance is seen, the setting relevant to anticipatory dialogue systems, de-escalation tools, and turn-taking agents. Forecasting is target-absent by construction: no feature may depend on utterance $n+1$. This constraint is easy to violate silently, e.g. via bidirectional encoders, dialogue-level pooling, or context windows not truncated at the decision point – and the architectures ERC has spent a decade refining are, almost without exception, built to violate it, because recognition never had to enforce it in the first place.

We ask a measurement question: how do conclusions change when the future is actually hidden and the learned model and transition baseline receive comparable current-state information? The two requirements are distinct. A causal model without gold labels should be compared with a baseline driven by train-only predicted labels; a baseline supplied with gold y_n should be compared with a model supplied with the same y_n. Our initial fixed-budget comparison confounded these regimes. The present audit corrects that design, tunes every learned family with the same validation budget, and propagates both dialogue and optimization uncertainty. The resulting picture is conditional rather than uniformly negative: deployable models consistently improve on the predicted-label baseline, while an information-matched oracle baseline remains hard on several corpora. Our contribution is a measurement framework with six parts:

1) a formal commitment point and target-absent data construction for next-turn emotion shift;
2) an executable eight-item leakage audit plus future-perturbation causality tests;
3) train-only predicted-label and gold-label transition baselines that expose the information available in each comparison;
4) equal four-configuration validation searches and checkpoint selection for six causal model families, evaluated over three seeds;
5) seed–dialogue hierarchical intervals, paired dialogue-cluster permutation tests, and a declared familywise Holm correction; and
6) provenance, future-access, and speaker-target controls that test whether conclusions survive feature re-extraction, target redefinition, and controlled leakage.

The final contribution matters because the two targets answer different questions. An interaction-level system may legitimately care whether the *conversation's* next turn changes emotion regardless of who speaks. A coaching or wellbeing system may instead care whether the *same person* changes state when they next speak. Reporting one as if it were the other can produce a semantically misleading claim even when the code is temporally leakage-free.

II. RELATED WORK

ERC backbones. Our four ERC-derived models are representative of three context-modeling paradigms: recurrent interlocutor-state tracking [7], relational graph convolution [8], [10], and directed relational attention [9]. Knowledge-enriched encoders [5], [11] and multimodal fusion architectures [14]–[18] inform our feature choices (COSMIC RoBERTa; MM-DFN [6]-style text+audio+visual). All are, in original form, bidirectional or graph-connected across the full dialogue – appropriate for recognition, but a direct leakage source if repurposed for forecasting without a causal mask; interactive memory [13] and contextual-reasoning networks [12] share this property.

Emotion forecasting (EFC). A parallel line explicitly targets an unobserved future utterance. Altarawneh et al. [20] formalize PEC as predicting turn $n+1$ from context strictly up to n, decomposing the signal into sequence, self-dependency, and recency terms (the strategy we re-implement as PEC-style, §V). Shi et al. [21], [22] model bilateral-speaker correlation and word-level cues in text+audio. Hi-EF [19] contributes a dedicated multimodal forecasting benchmark. MREPC [23] pre-trains contrastively over a short-term cross-speaker window. PUGCN [24] substitutes a pseudo-utterance-guided contrastive objective for the absent target (the strategy behind our Pseudo-future-style re-implementation). EDFC [25] targets a next-utterance emotion distribution; ERFC [26] shows feasibility in a deployed call-center setting. All are cross-speaker by design and, to our knowledge, none audits a leakage boundary as strict as ours. Notably EDFC and MM-DFN appeared at ICASSP [6], [25] and the EPC papers at Interspeech [21], [22], making this a speech-community concern as much as an NLP one.

Emotion shift and flip: adjacent, not forecasting. ESD-ERC [27], CFN-ESA [28], and EmoShiftNet [29] use shift labels to sharpen *recognition* of an observed current utterance – computed over turns the model can already see. Emotion-flip reasoning [30], [31] runs backward: given a flip that has already occurred, it attributes the past trigger. Neither predicts an unobserved future shift under a leakage-safe protocol; we exclude both, noting our transition matrix is close in spirit to the inertia signal they are designed to overcome.

Leakage and baseline validity. Kapoor and Narayanan [32] survey at least 294 papers across 17 fields and find leakage the dominant cause of irreproducible ML-based findings, proposing the taxonomy our 8-item audit is modeled after. Ferrari Dacrema et al. [33] show most neural recommenders fail to beat tuned classical baselines once evaluation is corrected; Lin [34] makes the same point for neural ranking. We view our transition-matrix result as the same phenomenon in emotion forecasting.

III. TASK DEFINITION AND THREAT MODEL

Let a dialogue be $D = \{(x_t, y_t, s_t)\}_{t=1}^{T}$, where x_t is an utterance-level feature vector, $y_t \in \{1, \ldots, K\}$ an emotion label, and s_t a speaker identifier. At commitment point n, a legitimate forecaster may use only

$$\mathcal{I}_n = \{x_{1:n}, s_{1:n}\} \quad \text{and, when declared, } y_n, \tag{1}$$

but no quantity computed from $x_{n+1:T}$ or $y_{n+1:T}$. This definition includes an important feature-provenance condition: x_t must itself be utterance-local. Merely slicing a dialogue-level bidirectional representation at index n does not make it causal.

A. TWO FORECASTING TARGETS

Our original and primary target is the *interaction-level immediate shift*

$$z_n^{\text{int}} = \mathbb{1}[y_{n+1} \neq y_n], \qquad n < T. \tag{2}$$

It is operationally appropriate when the question is "will the emotion expressed in the next turn differ from the emotion expressed now?" It is agnostic to speaker identity and therefore mixes two strata: $s_{n+1} = s_n$ and $s_{n+1} \neq s_n$.

For semantic clarity we additionally define the *next-own-utterance self-shift*. Let

$$g(n) = \min\{m > n : s_m = s_n\} \tag{3}$$

when such an utterance exists. Then

$$z_n^{\text{self}} = \mathbb{1}[y_{g(n)} \neq y_n]. \tag{4}$$

This target asks whether the current speaker changes emotion by the time they next speak, allowing intervening turns by others. It removes the immediate cross-speaker label comparison, but it is not "more correct" in the abstract: it answers a different product and scientific question, has a variable forecast horizon, and excludes terminal appearances of each speaker. We report both rather than silently substituting one for the other.

B. WHAT COUNTS AS LEAKAGE?

Table 1 distinguishes information leakage from other sources of optimistic or misleading evidence. The first three rows violate Eq. (1); the remaining rows do not necessarily leak the future but can still make comparisons invalid. This separation prevents the term "leakage" from becoming a catch-all diagnosis.

IV. PROTOCOL

Commitment point. For decision point n, all input features (text/audio/visual) are restricted to utterances $\leq n$; the target label is $\mathbb{1}[y_{n+1} \neq y_n]$. All six learned models (§V) are structurally causal, not causal-by-convention.

Information-matched regimes. In the *deployable* regime, learned models receive utterance features and causal history but no gold emotion label. The transition baseline instead uses $\hat{y}_n$ from a linear current-emotion classifier fit exclusively on training utterances; its regularization is selected on the validation fold. In the *oracle diagnostic*, the gold one-hot y_n is appended to every learned model and the transition baseline conditions on the same gold y_n. This is not a deployment claim: it isolates the value of forecasting beyond a shared, perfect observation of current state. Comparing a no-label model only against a gold-label transition matrix is retained as a useful stress test, but is not used as the deployable headline.

Causal construction and a zero-tolerance check. DialogueRNN's global/party/emotion states update by chained `GRUCell` calls that never touch $x_{t+1:}$. DialogueGCN uses *windowed* causal edges ($j \to i$ iff $j \leq i$ and $i - j \leq 6$); DAG-ERC uses *unbounded* causal attention ($j \leq i$, any distance) – two families differing sharply in receptive field yet still landing within the same $\lesssim$0.04 AUC band (§V), which strengthens rather than weakens Finding 4. We verified causality directly rather than trusting architecture alone: perturbing every input at $t \geq n$ to random noise leaves outputs at $t < n$ changed by exactly 0.0 (floating-point identical) for every model.

Automated leakage audit. Every experiment emits an 8-item checklist: last-utterance masking, model causality, per-utterance feature scope, speaker-split disjointness, val-only threshold tuning, dialogue-level bootstrap, train-fold-only baseline fitting, and baseline completeness.

Padding and supervision masks. Dialogue batches are right padded, but the final true utterance of every dialogue and every padded position receive an ignore label. Classification loss, threshold selection, and metrics operate only on valid decision points. For the corrected Pseudo-future implementation, its auxiliary next-embedding regression is masked by the same valid decision-point tensor. This detail matters because unmasked padding creates a length-dependent training signal even though it does not directly expose the future.

Strong baseline. Beyond base-rate and no-change/inertia, we fit a speaker×emotion transition matrix $P(y_{n+1} \mid y_n, \text{speaker})$ on the train fold only (Laplace α=1.0; a speaker's row is used only with $\geq$20 training observations, else global backoff). This is the baseline every model must beat.

Equal validation budget. Each architecture receives the same four candidates spanning hidden size $\{64, 128, 256\}$, learning rate $\{3 \times 10^{-4}, 10^{-3}\}$, dropout $\{.1, .3\}$, and focal versus class-balanced cross-entropy loss. Seed 0 is used for configuration selection on validation ROC–AUC only. The selected configuration is then trained with seeds $\{0, 1, 2\}$ for at most 30 epochs; the validation-selected checkpoint uses a five-epoch patience after at least five epochs. The test fold is never evaluated during search or checkpoint selection. A validation-selected binary threshold affects only thresholded metrics, not ROC–AUC, PR–AUC, Brier score, or calibration error.

Metrics. ROC–AUC is primary because it is threshold-free. We additionally report PR–AUC for the imbalanced shift class, Brier score and expected calibration error, shift-F1, precision, recall, and balanced accuracy. Predictors assigning more than 98% of decisions to one class are automatically flagged. PR–AUC is interpreted relative to each corpus's shift prevalence, not compared naively across corpora [35]; calibration is measured on probabilities, separately from discrimination [36].

Significance. Differences are estimated with a hierarchical bootstrap [37] that first samples training seeds and then resamples whole test dialogues (1,999 replicates). Two-sided paired dialogue-cluster permutation tests use 4,999 sign flips and the plus-one correction, so a reported p-value is never zero. Holm–Bonferroni correction is applied jointly across the two information regimes, six models, and six dataset-configurations (72 declared comparisons).

Reproducibility contract. Random seeds are fixed at 0, 1, and 2. Each dataset is loaded and split before model construction; no fitted current-emotion classifier or cached prediction is shared across seeds. Raw scores for every seed are retained and jointly enter inference. Dataset manifests record SHA-256 hashes, split identifiers are deterministically ordered, and independent tests verify that perturbing inputs at and after the forecast boundary cannot alter earlier logits. DailyDialog validation/test dialogues exactly duplicated in training are

TABLE 1. Threat model for conversational emotion forecasting. The audit treats target-derived features and target-derived model states as zero-tolerance failures; the other threats require separate statistical or reporting controls.

Threat	Failure mechanism	Detection	Required control
Target-turn exposure	x_{n+1} enters a window, pooled context, or auxiliary input	inspect window boundaries; sweep visible future turns	truncate at n before any aggregation
Bidirectional state	output at n depends on representations from $n+1:T$	perturb future inputs and compare prefix logits	unidirectional recurrence or explicit causal mask
Feature provenance	nominally per-turn x_n was extracted with dialogue-level future context	trace the feature extractor, not only the classifier	utterance-local extraction or causal re-extraction
Split contamination	duplicate dialogue text or the same global speaker occurs across folds	hash dialogue text; audit global identities	group-aware split and duplicate removal
Selection leakage	test fold determines threshold, epoch, or model family	inspect tuning log	validation-only selection with fixed test evaluation
Metric degeneracy	high base rate makes always-shift F1 look competitive	report prediction rate and balanced metrics	AUC, balanced accuracy, and constant baselines
Semantic target drift	cross-speaker next-turn change is described as same-person emotional change	stratify by $s_{n+1} = s_n$; evaluate Eq. (4)	name the target and deployment contract

removed before any search.

V. EXPERIMENTS

Data. The primary analysis uses four text corpora with released RoBERTa features (IEMOCAP [2], MELD [1], EmoryNLP [3], DailyDialog [4]; COSMIC pipeline [5]), plus two multimodal configurations (IEMOCAP, MELD with MM-DFN [6] text+OpenSmile-audio+DenseNet-visual features) are reported only as feature-source sensitivity analyses because the available release does not include the complete upstream extraction code. Table 2 gives size and shift-rate statistics (0.20–0.72) together with an F1-degeneracy analysis referenced below.

The six configurations are not six independent corpora: IEMOCAP and MELD each appear once with COSMIC text features and once with MM-DFN multimodal features. This paired reuse is intentional for the feature-source comparison, but inferential claims are made per configuration and are not treated as six independent replications. COSMIC features are 1024-dimensional RoBERTa vectors; MM-DFN provides 600/100-dimensional text, 300/1582-dimensional audio, and 342-dimensional visual vectors for MELD/IEMOCAP, respectively. We use official or released train/test identifiers; MM-DFN lacks validation identifiers, so a deterministic 10% slice of the training dialogues is reserved for validation. We verified the released COSMIC producer scripts for IEMOCAP, MELD, and EmoryNLP and confirmed that the stored RoBERTa rows are generated per utterance. As an independent control that does not rely on released embeddings, we also fit TF–IDF vocabulary/IDF and truncated SVD on training utterances only and transform every utterance independently. DailyDialog contains 134 test and 101 validation dialogues exactly duplicated in training; all revised DailyDialog results remove those dialogues before configuration selection or evaluation.

A. COMPARED MODELS

The benchmark spans recurrence, structured speaker state, graph propagation, directed attention, and two forecasting-specific strategies. To keep the question focused on the temporal contract, each model maps the same feature sequence to one logit per observed turn and uses the same loss and optimization budget. DialogueRNN maintains global and party states; causal DialogueGCN propagates messages only from earlier nodes in a six-turn window; DAG-ERC uses unbounded causal attention; PEC-style concatenates sequence, self-dependency, and exponentially weighted recency states; and Pseudo-future-style predicts an embedding for x_{n+1} from the causal state before classifying shift. The actual x_{n+1} is only an auxiliary regression target during training and never a forward-path input.

State-factorized control. Shift-only supervision collapses every directed emotion transition to one bit. Motivated by work that predicts the next-utterance emotion distribution directly [25], we add a causal multi-task control with three heads: current emotion, future emotion, and direct shift. Its factorized shift probability is $1-\sum_k p(y_n=k \mid \mathcal{I}_n)p(y_{n+1}=k \mid \mathcal{I}_n)$; a convex blend with the direct head is selected on validation data only. No gold label or future utterance enters inference. Because this model was designed after the audit, it is evaluated in a separately declared four-corpus family rather than inserted post hoc into the 72 main comparisons. It reaches .628/.628/.546/.747 ROC–AUC on IEMOCAP/MELD/EmoryNLP/DailyDialog and improves over the deployable transition baseline by +.052/+.041/+.043/+.166; MELD and DailyDialog remain significant after four-test Holm correction ($p = .0036$ and .0008). Its score is comparable to, rather than higher than, the best main-benchmark architecture.

We also retain a bidirectional GRU solely as a positive leakage control. It is not a candidate forecaster. Its purpose is diagnostic: if the safe and leaky models perform identically, either the target utterance carries little useful signal or the benchmark implementation is failing to learn; if the leaky model improves sharply, the latter explanation becomes less plausible.

B. SPEAKER-RELATION AUDIT

Table 3 exposes a compositional property hidden by aggregate shift rates. In the test folds, only 27.8% of IEMOCAP and 24.2% of MELD adjacent decisions retain the same speaker;

TABLE 2. Dataset statistics and F1 degeneracy. Always-shift F1 = $2r/(1+r)$ for shift rate r (analytic; always-shift balanced accuracy is 0.500 by construction, omitted). Transition-BA ≈0.50 on MELD/EmoryNLP/IEMOCAP-mm shows standard F1 sits at the degenerate ceiling on exactly the corpora where shift is common; only IEMOCAP and DailyDialog clear it.

Dataset	#Dial	#Utt	Shift r	Always-shift F1	Transition BA
IEMOCAP	151	7,433	0.429	0.600	0.611
MELD	1,432	13,708	0.581	0.735	0.500*
EmoryNLP	827	9,489	0.722	0.839	0.500*
DailyDialog	13,118	102,979	0.198	0.331	0.681
IEMOCAP-mm	151	7,433	0.429	0.600	0.500*
MELD-mm	1,432	13,708	0.581	0.735	0.500*

EmoryNLP retains the speaker in 9.1%, and DailyDialog alternates speakers by construction in the released feature file. Thus, most labels in Eq. (2) compare the emotions of two different people.

This composition is consequential but dataset-dependent. IEMOCAP's cross-speaker shift rate is 0.509 versus 0.222 for same-speaker adjacent turns; MELD's is 0.631 versus 0.423. In EmoryNLP the rates are similar (0.719 versus 0.744), although its same-speaker stratum contains only 82 test points and should not support a strong difference claim. DailyDialog has no immediate same-speaker stratum. Consequently, speaker switching is a strong correlate of the label in IEMOCAP and MELD, not a universal explanation for all results.

A 5,000-sample dialogue-cluster bootstrap confirms that the switch-minus-same shift-rate difference is not merely an utterance-count artifact on IEMOCAP ($+0.287$, 95% CI $[+0.200, +0.373]$) or MELD ($+0.208$, $[+0.151, +0.265]$); both two-sided bootstrap $p < .001$. EmoryNLP shows no supported difference (-0.025, $[-0.119, +0.072]$, $p = .614$). DailyDialog cannot support this contrast because its same-speaker cell is empty. Resampling whole dialogues is important here: treating 6,740 DailyDialog utterance pairs or 2,330 MELD pairs as independent would understate clustered uncertainty.

The self-shift target also changes the effective sample and horizon. It removes the final appearance of every speaker, leaving 80.0–98.1% as many test decisions as the immediate target, and its mean gap ranges from 1.98 to 2.51 turns. The shift-rate change is largest on IEMOCAP (0.429 immediate to 0.263 self), confirming that the original aggregate rate partly reflects cross-speaker contrast rather than within-person instability.

Table 4 shows that inertia remains predictive after redefining the target, but its strength decreases on three corpora. Speaker conditioning contributes exactly zero AUC on every corpus in this evaluation: IEMOCAP's global session-and-actor identities are genuinely unseen at test time, and MELD, EmoryNLP, and DailyDialog roles are qualified by dialogue ID, so every test-time speaker key is, by construction, absent from the train-fold speaker table and the estimator always backs off to the global row. The baseline's full strength therefore comes from the emotion-specific transition alone; this evaluation provides no evidence that speaker-identity conditioning helps, only that dialogue-local role identities cannot be tested for a within-person effect without a genuinely cross-dialogue speaker identifier.

Forecast horizon is another hidden source of heterogeneity. Table 5 groups self-shift decisions by the number of turns until the current speaker returns. IEMOCAP and MELD show markedly higher shift rates after gaps of at least three turns than at gap one; EmoryNLP is non-monotonic, and DailyDialog contains only gap-two decisions. A single aggregate self-shift score therefore combines different horizons just as the immediate target combines different speaker relations. Future benchmarks should either report horizon strata or define a fixed time/turn horizon prospectively.

The self-shift benchmark supports a bounded generalization. DailyDialog remains the stable positive case at the longer, exactly two-turn horizon ($+.114$ to $+.129$ AUC versus Trans-Pred). IEMOCAP models are directionally positive but the test split contains only 31 dialogue clusters; MELD gains are small, and EmoryNLP is tied. Redefining the target therefore changes both prevalence and model evidence rather than merely renaming the original task.

Main result (Table 7). We evaluate a plain GRU, causal re-implementations of DialogueRNN [7], DialogueGCN [8], and DAG-ERC [9], plus PEC-style [20] and pseudo-future-style forecasting strategies [24]. Information matching changes the conclusion. In the deployable regime, all six models exceed the train-only predicted-label transition baseline on every primary text corpus in point estimate. The gain is decisive on duplicate-free DailyDialog ($+0.160$ to $+0.167$ ROC–AUC; all adjusted $p = .0144$) and smaller elsewhere. Among the primary text corpora, DialogueRNN on MELD is the only other comparison that survives the 72-test familywise correction ($\Delta = +0.041$, adjusted $p = .0352$); all six MELD-mm gains also survive but sit in a feature-source sensitivity configuration rather than a primary corpus. Positive but uncorrected differences on IEMOCAP and EmoryNLP are evidence of possible benefit, not conclusive wins.

The oracle diagnostic asks a different question. When both sides receive gold y_n, all six models remain clearly better on DailyDialog ($+0.106$ to $+0.132$; adjusted $p = .0144$), whereas the transition baseline is stronger on MELD (-0.032 to -0.064), with four model differences surviving correction. IEMOCAP is effectively tied and EmoryNLP is directionally favorable to the baseline. Thus, the transition matrix is neither a universal lower bound nor a straw baseline: its strength depends on whether current emotion is known and on corpus-specific transition structure.

Scope. Four systems are causal, strategy-level re-implementations of representative ERC families; PEC-style and pseudo-future-style are forecasting-inspired strategies. Our claims concern evaluation practice and model families, not reproduction of any individual published score. The MM-DFN feature release is a sensitivity analysis because its complete upstream producers are not available; primary claims rest on four text corpora and are repeated using independently fitted, strictly utterance-local text features. Finally, structural causality is verified rather than inferred from architecture names: perturbing all inputs beyond the commitment point

TABLE 3. Speaker relation and target composition on the four unique test corpora. Same-next is $P(s_{n+1} = s_n)$. Self-shift uses the next-own-utterance target in Eq. (4); gap is the mean number of turns from n to $g(n)$. IEMOCAP/MELD multimodal configurations have the same test labels and are therefore not duplicated.

Corpus	Immediate N	Same-next	Shift|same	Shift|switch	Self N	Self shift	Mean gap
IEMOCAP	1,592	0.278	0.222	0.509	1,561	0.263	1.98
MELD	2,330	0.242	0.423	0.631	1,864	0.538	2.21
EmoryNLP	905	0.091	0.744	0.719	739	0.673	2.51
DailyDialog	6,740	0.000	–	0.198	5,740	0.180	2.00

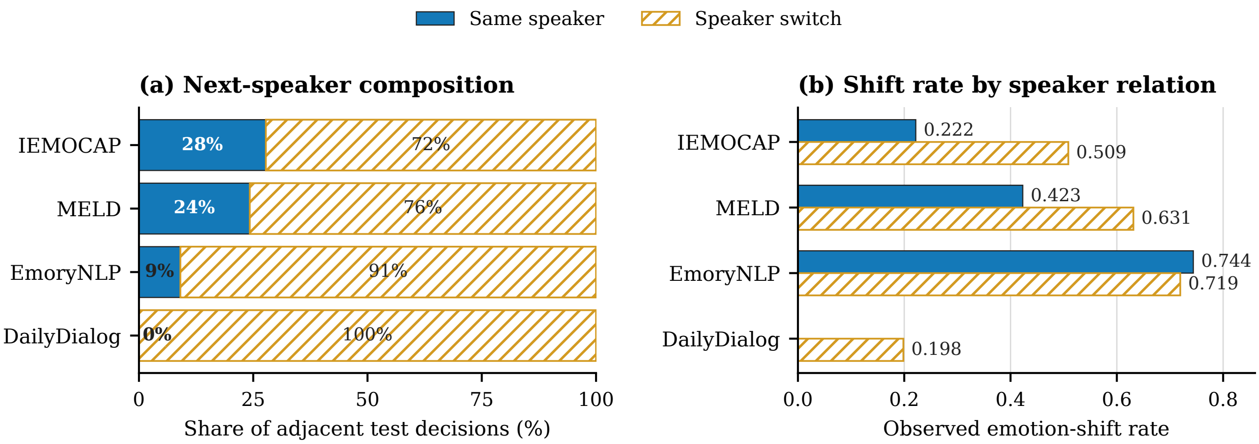

FIGURE 1. Speaker composition and emotion-shift rate at adjacent test decision points. Filled blue bars denote same-speaker transitions; hatched bars denote speaker switches, preserving the distinction in grayscale. DailyDialog has no adjacent same-speaker decisions. Exact denominators are given in Table 3.

TABLE 4. Train-fold transition baselines under the two targets (test AUC). Global conditions only on current emotion; speaker conditions on current emotion and speaker with sparse/unseen-speaker backoff. These are label-only diagnostics, not learned-model results.

	Immediate		Self-shift	
Corpus	Global	Speaker	Global	Speaker
IEMOCAP	0.642	0.642	0.586	0.586
MELD	0.676	0.676	0.663	0.663
EmoryNLP	0.608	0.608	0.592	0.592
DailyDialog	0.691	0.691	0.657	0.657

TABLE 5. Self-shift composition by forecast gap. Cells show test N / shift rate; – indicates that the released test split contains no decision at that gap.

Corpus	Gap 1	Gap 2	Gap ≥ 3
IEMOCAP	442 / .222	864 / .234	255 / .431
MELD	565 / .423	961 / .580	338 / .612
EmoryNLP	82 / .744	464 / .619	193 / .772
DailyDialog	–	5,740 / .180	–

changes every earlier logit by exactly zero for every tested implementation.

Tables 7 and 8 average three independently trained seeds; inference uses all three raw score vectors rather than a single representative run. The strongest primary result is therefore not merely numerical: on DailyDialog every model has a positive delta in every seed, the 95% hierarchical intervals exclude zero, and all paired permutation tests remain significant after the declared joint correction.

Implementation corrections. The audit found and repaired two defects before the revised experiment was declared: PEC recency weights were reversed, and pseudo-future auxiliary MSE included padded positions. Separate cache namespaces retain historical runs for traceability, but Tables 7–8, all inference, and all supplementary controls use only the corrected implementations.

Independent utterance-local features. Table 9 removes dependence on released deep embeddings. TF–IDF vocabulary and IDF, followed by rank-128 truncated SVD, are fitted only on training utterances; each utterance is then transformed independently with finite-value checks. The local-feature GRU remains substantially above its matched predicted-label transition baseline on IEMOCAP and DailyDialog and directionally above it on EmoryNLP; MELD is tied. Absolute AUC is lower than with RoBERTa on three corpora, as expected for a shallow representation. Crucially, the DailyDialog advantage is unchanged in magnitude (+.164 versus +.166 with RoBERTa), so its headline improvement cannot be attributed solely to unobserved context in an upstream encoder. The mixed pattern also cautions against treating any released feature source as architecture-neutral.

Same-architecture dose response (Fig. 2). A single-layer Transformer holds parameters, positional encoding, pooling, optimizer, and checkpoint rule fixed; only its attention mask changes so that position n sees $k \in \{0, 1, 2, 4, \text{all}\}$ future utterances. A single layer is required for k to mean what the sweep claims: with a stacked multi-layer encoder using the same window mask at every layer, a later layer's query at n can read an earlier layer's output at $n + k$, which already mixed in x_{n+2k}, giving an effective receptive field of $\sim Lk$

TABLE 6. Complete next-own-utterance self-shift ROC–AUC (three-seed mean). The full causal dialogue history is retained; only the supervised target index changes. Trans-Pred is information-matched. DailyDialog gains all survive the separate 24-comparison Holm correction (p_{adj} = .0048); other corrected comparisons are not significant.

Corpus	Trans-Pred	GRU	DRNN	DGCN	DAG-ERC	PEC	PseudoFut
IEMOCAP	.541	.609	.602	.597	.598	.593	.605
MELD	.584	.593	.600	.608	.610	.593	.593
EmoryNLP	.518	.509	.503	.514	.515	.501	.523
DailyDialog	.577	**.699**	**.706**	**.698**	**.691**	**.695**	**.693**

TABLE 7. Deployable shift ROC–AUC (three-seed mean). Trans-Pred uses current-emotion predictions from a train-only linear classifier; learned models receive no gold emotion. Bold values exceed the matched baseline. DailyDialog is duplicate-free. MM rows are feature-source sensitivity analyses, not independent primary corpora. Inference uses a seed–dialogue hierarchical bootstrap and paired dialogue-cluster permutation; Holm correction spans 72 declared deployable+oracle comparisons. All six DailyDialog gains, all six MELD-mm gains, and DialogueRNN's MELD gain survive correction.

Dataset	Trans-Pred	GRU	DRNN	DGCN	DAG-ERC	PEC	PseudoFut
IEMOCAP	.576	**.647**	**.627**	**.609**	**.607**	**.632**	**.631**
MELD	.588	**.621**	**.629**	**.623**	**.608**	**.620**	**.600**
EmoryNLP	.501	**.521**	**.533**	**.536**	**.554**	**.539**	**.542**
DailyDialog	.580	**.746**	**.744**	**.748**	**.744**	**.743**	**.741**
IEMOCAP-mm	.566	.560	**.598**	**.593**	.559	**.581**	**.578**
MELD-mm	.584	**.723**	**.716**	**.724**	**.720**	**.720**	**.723**

TABLE 8. Oracle information-matched shift ROC–AUC (three-seed mean). Both the transition baseline and every learned model receive gold current emotion. Bold values exceed Trans-Gold. DailyDialog and MELD-mm gains survive joint Holm correction; on MELD, four of the six negative differences survive.

Dataset	Trans-Gold	GRU	DRNN	DGCN	DAG-ERC	PEC	PseudoFut
IEMOCAP	.642	**.649**	.615	.617	.616	.631	.610
MELD	.676	.630	.629	.628	.644	.632	.612
EmoryNLP	.608	.553	.561	.561	.563	.591	.525
DailyDialog	.694	**.808**	**.820**	**.826**	**.822**	**.804**	**.800**
IEMOCAP-mm	.642	.566	.590	.598	.577	.594	.569
MELD-mm	.676	**.802**	**.791**	**.805**	**.807**	**.798**	**.798**

TABLE 9. Strict utterance-local feature control (three-seed ROC–AUC). Both the GRU features and the baseline's current-emotion classifier are refit from the same train-only TF–IDF/SVD rows. Intervals and p-values use the same seed–dialogue procedures as the main experiment.

Corpus	GRU	Trans-Pred	Δ [95% CI]
IEMOCAP	.642	.552	+.089 [.010,.154]
MELD	.539	.538	+.001 [-.027,.028]
EmoryNLP	.555	.494	+.061 [.014,.112]
DailyDialog	.673	.509	+.164 [.141,.185]

for L layers rather than k. The response is not monotone. MELD rises from .620 at $k = 0$ to .685 at $k = 1$, then falls to .651/.637/.632. Duplicate-free DailyDialog similarly rises from .736 to .780 and falls to .774/.758/.751. The other four configurations vary by at most .022 without a consistent direction. Thus, exposure to the target utterance can materially inflate performance, but additional future context need not add signal and may dilute or complicate optimization. Because architecture is held fixed and the receptive field is exactly k, this conclusion no longer relies on comparing a look-ahead GRU with a bidirectional GRU, nor on a multi-layer encoder whose effective receptive field exceeds its nominal window.

C. SPLIT AND LEAKAGE ROBUSTNESS

Exact dialogue contamination. We hashed each dialogue by its complete ordered utterance text and compared validation/test hashes against the training fold (Table 10). IEMOCAP and EmoryNLP contain no exact cross-fold matches; MELD contains one duplicate test dialogue; DailyDialog contains 134/1,000 duplicate test dialogues and 101/1,000 duplicate validation dialogues. This is a corpus property visible before model fitting, not a model-specific leak, but it can make a positive result look stronger.

TABLE 10. Exact train-fold dialogue duplicates. A duplicate has an identical full ordered text sequence, not merely an overlapping utterance.

Corpus	Test N	Test Dup.	Validation N	Validation Dup.
IEMOCAP	31	0	12	0
MELD	280	1	114	0
EmoryNLP	79	0	89	0
DailyDialog	1,000	134	1,000	101

We remove every matching DailyDialog validation/test dialogue before any configuration search, checkpoint selection, baseline fitting, or reported evaluation. Test dialogues fall from 1,000 to 866 and validation dialogues from 1,000 to 899. All DailyDialog results in this paper—main, oracle, state-factorized, local-feature, self-shift, and dose-response—use this same clean split.

Earlier unequal-architecture look-ahead controls are retained in the repository for historical traceability but are not used for the revised leakage claim; Fig. 2 supersedes them by holding architecture and parameter count fixed.

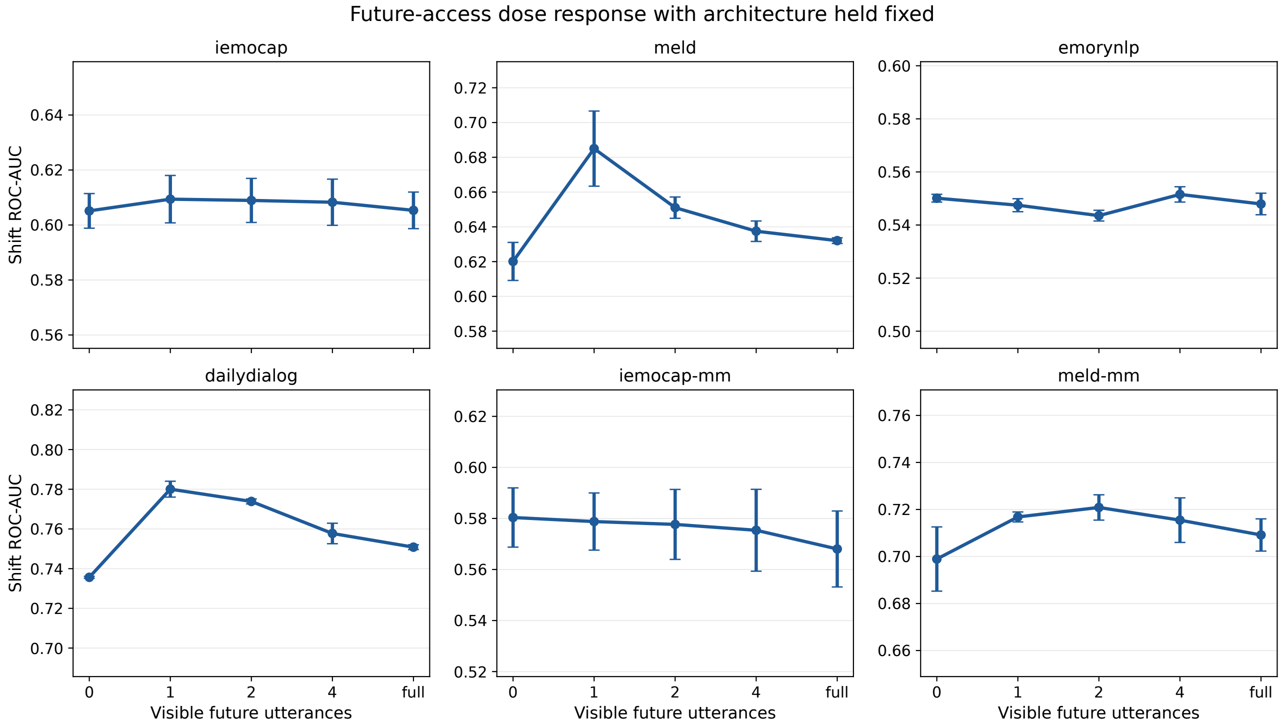

FIGURE 2. Future-access dose response with one fixed Transformer architecture. Error bars show three-seed standard deviation. MELD and DailyDialog peak when the target utterance alone becomes visible; none of the six configurations supports a general monotonic-more-future claim.

VI. FINDINGS AND DISCUSSION

The revised evidence supports six bounded findings.

1) Information matching is consequential. No-label models consistently exceed a baseline fed noisy train-only label predictions in point estimate, whereas a gold-label transition matrix is competitive or stronger on several corpora.
2) DailyDialog is the most stable positive case: every tested model beats both matched baselines after duplicate removal, across every seed, with corrected-significant hierarchical inference.
3) MELD is conditional: learned models improve on the deployable predicted-label baseline, but most lose to the oracle gold-label baseline. The current-state observation, not only architecture, changes the answer.
4) Ranking and probability quality diverge. PR–AUC corroborates DailyDialog discrimination, while Brier/ECE show that learned scores require calibration before risk interpretation.
5) Feature provenance remains part of the estimand. The independently fitted local-feature control preserves large DailyDialog and IEMOCAP gains, but not the MELD gain.
6) Adjacent-turn shift is predominantly cross-speaker in all four corpora; interaction-level next-turn shift and a person's next-own-utterance self-shift are distinct targets and are evaluated separately here.

These findings do not establish that one architecture class is generally superior or irrelevant. They establish a narrower measurement result: the sign and magnitude of model-versus-transition differences depend on the information supplied to both sides, and conclusions that omit this factor can reverse. Architecture spreads within a configuration are often modest, but the equal search is intentionally limited and the systems are causal re-implementations. Larger searches or new forecasting-native systems may change their ordering.

The multimodal rows reinforce the need for restraint. Their point estimates can be large, but the complete upstream producers are unavailable and repeated identical MELD strings do not always map to identical released vectors. They are therefore useful sensitivity evidence, not the basis for a claim that a modality or representation is intrinsically better. The independently generated local control supplies the stronger provenance test.

A. WHAT THE SPEAKER AUDIT CHANGES

The audit in §V-B changes the language appropriate for the task. The immediate target is not "will this person change emotion on their next turn?" unless evaluation is restricted to $s_{n+1} = s_n$. It is "will the emotion expressed by the conversation's next utterance differ from the current utterance?" This remains meaningful for a dialogue manager that cares about the next observed affect regardless of speaker, but it combines within-person dynamics with cross-person response dynamics.

The distinction also clarifies the transition result. A model can exploit emotion persistence, common cross-speaker response pairs, or a correlation between turn-taking and label change. The last signal is available at commitment time only if the next speaker is scheduled or otherwise known. Our headline models are not given s_{n+1}, so the stratified rates are explanatory diagnostics rather than an oracle input. Future benchmarks should state whether next-speaker identity is known and, if so, provide it symmetrically to learned models and baselines.

B. OPERATIONAL IMPLICATIONS

Temporal validity is necessary but not sufficient for deployment. A high-AUC alert can still be poorly calibrated, too late for intervention, or unfairly concentrated on particular speakers. The commitment point must correspond to a real action window: after the current turn is available but before any acoustic, lexical, or metadata trace of the next turn enters the system. Production logs should retain feature timestamps so this boundary remains auditable.

Thresholds should follow intervention costs rather than F1 alone. In de-escalation, a false negative may miss harm while repeated false positives create alert fatigue. AUC supports model comparison but does not select an operating point; calibration, precision–recall tradeoffs, and cost-sensitive validation are needed before use. This distinction is visible in the strongest ranking result: duplicate-free DailyDialog DialogueGCN reaches ROC–AUC .748 and PR–AUC .441, but its uncalibrated Brier score/ECE are .188/.224, versus .153/.052 for the weaker predicted-label transition baseline. Post-hoc calibration must therefore be fitted on held-out deployment-like data before probabilities are interpreted as risks. Emotion labels also concern sensitive and culturally ambiguous human states. This work audits measurement; it does not establish that an automated emotional intervention is safe or appropriate.

C. THREATS TO VALIDITY AND LIMITATIONS

Construct validity. Emotion annotations are discrete expressions, not direct measures of internal affect. Both targets inherit annotator disagreement and collapse different transition types into one event. The self-shift benchmark is complete across the six tested model families, but it changes both the target horizon and sample composition and should not be read as a direct ranking against immediate-shift scores.

Internal validity. Causality tests cover model forward paths, masks, and provided feature arrays, but cannot retroactively certify every upstream feature-extraction step. COSMIC producer code directly supports utterance-local extraction for three corpora; DailyDialog is supported by schema/output checks, while MM-DFN is not certified to the same level. The train-only TF–IDF/SVD control reduces, but cannot remove, construct limitations from discrete emotion labels and splits.

Split validity. We remove exact train–test dialogue duplication from every DailyDialog headline and supplementary run, but not subtler writing-style leakage. MELD, EmoryNLP, and DailyDialog identities in the released pickles are dialogue-local, not verified global persons. Only IEMOCAP's session split provides an unseen-actor test. Speaker conditioning elsewhere means local role conditioning, not person-level generalization.

Statistical validity. Three seeds characterize optimization variability but do not make the six configurations independent: two corpora recur under different feature sources. Joint Holm correction is intentionally conservative. Hierarchical intervals and paired permutation tests use all seeds and whole-dialogue clusters, but neither procedure replaces replication on a prospectively designed forecasting corpus.

Model scope. Forecasting-specific systems are strategy-level re-implementations because original code was unavailable; we do not claim every detail of a named publication. The common four-configuration search favors comparability over per-model best-case tuning and does not span architecture-specific graph windows or all loss families. Any larger search should apply the same budget to competing families and retain validation-only selection.

Information matching is imperfect in both directions. In the oracle diagnostic, the gold one-hot label is concatenated to the per-turn feature vector at every position $t \leq n$, so a sequential or attention-based model's representation at n is conditioned on the entire observed label history $y_1, \ldots, y_n$, not on y_n alone; the transition baseline, by construction, conditions only on the single most recent label. "Information-matched" therefore means matched at the single quantity the title foregrounds – current-emotion availability – not matched in the total label information reaching each side, and the oracle gap likely understates the transition baseline's competitiveness under a strict single-label match. Symmetrically, the deployable regime's transition baseline consumes a hard current-label prediction from a linear classifier rather than the classifier's posterior, discarding calibrated uncertainty that a posterior-marginalized or cross-fitted baseline could exploit; this is a conservative choice for our headline claim (a stronger predicted-label baseline could only narrow, not widen, the learned-model advantage we report), but it means the deployable comparison is not the strongest baseline we could construct. Both limitations bound the oracle and deployable comparisons in opposite directions and are left as reporting caveats rather than resolved with a new baseline family in this submission.

VII. REPRODUCIBILITY ARTIFACTS

The repository is organized so each major claim has an executable producer rather than being a manually assembled table. Table 11 maps the paper's evidence to source and output artifacts. Feature pickles are not redistributed because of size and upstream licensing; the README records expected filenames and provenance. Every result script is resumable through dataset-specific JSON caches, and historical caches are kept separate from revised experiment caches.

The minimum reproduction sequence is to place the released features under `data/feat`, run the data-quality and causality modules, and then run the primary revision module; exact commands are in the packaged README. Supplementary producers use isolated cache namespaces and resume

TABLE 11. Claim-to-artifact map. Producers are under `src/`; reviewable outputs are under `results/`.

Claim	Producer	Reviewable output
Main benchmark	`access_revision_experiments`	JSON/Markdown + raw scores
Structural causality	`check_causality`	causality check
Uniform leakage dose	`access_uniform_dose_response`	figure + JSON
Joint inference	`evaluate`, `holm_correction`	72 corrected tests
Speaker audit	`access_speaker_analysis`	target/stratum JSON
Full self-shift	`access_self_shift_benchmark`	six-model raw scores
Local feature control	`access_local_feature_control`	train-only features/results
Data/provenance audit	`access_data_quality`	alignment/integrity report

from completed seeds. The paper source includes the IEEE Access class, fonts, figures, and a compiled PDF matching the submitted LaTeX source.

VIII. MINIMAL REPORTING CHECKLIST

Standalone checklist for future EFC-forecasting papers: (1) state the commitment point and verify no feature crosses it; (2) name the semantic target—interaction-level next turn or same-person next own turn—and report speaker-relation composition; (3) verify causality directly (e.g. a future-perturbation test), not by architecture claim alone; (4) trace feature provenance upstream of the classifier; (5) match current-label information between learned models and train-only transition baselines; (6) report ROC–AUC, PR–AUC, calibration, balanced accuracy, and prediction rate rather than F1 alone; (7) use seed–dialogue-aware paired inference corrected jointly across the declared comparison family; (8) label leaky ablations and exclude them from headline comparisons; (9) check duplicate dialogues and global speaker overlap; and (10) release masks, split identifiers, raw seed-level predictions, and audit logs.

IX. CONCLUSION

Information-matched evaluation changes the interpretation of conversational emotion-shift forecasting. Models that receive no gold current label can meaningfully outperform a transition baseline driven by train-only predicted labels; on duplicate-free DailyDialog this conclusion is large, seed-consistent, and familywise significant. A transition matrix supplied with gold current emotion is nevertheless competitive on IEMOCAP and stronger on MELD, showing why privileged-label diagnostics must not be presented as deployable comparisons. Independent local features preserve the DailyDialog result, while calibration metrics warn that strong ranking is not yet reliable risk estimation. Finally, adjacent interaction shift and same-person self-shift are both legitimate but different targets requiring explicit horizons and matched baselines. The general lesson is methodological rather than architectural: state the temporal, information, provenance, and semantic contracts before interpreting a model comparison [32]–[34]. The submission supplement contains executable producers, locked environment files, checksums, and raw seed predictions. The public project page is https://github.com/GawaineXiukkie/emotion-forecasting-leakage-audit; an immutable release tag will be created from the final reviewed artifact before the reproducibility claim is made public.

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